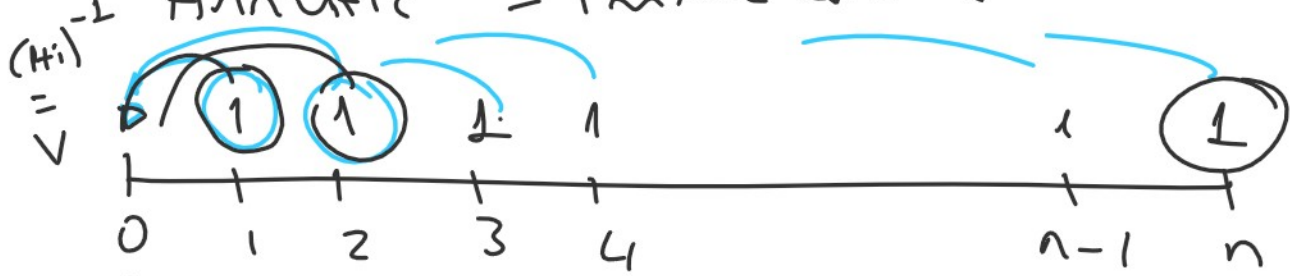


Annuité - immediate



$1 + r + r^2 + r^3 + \dots + r^{n-1} = \frac{1 - r^n}{1 - r}$ geometric series
r: common ratio.

$$\begin{aligned}
 a_{\overline{n}|} &= 1 \cdot v + 1 \cdot v^2 + 1 \cdot v^3 + \dots + 1 \cdot v^n \\
 &= v (1 + v + \dots + v^{n-1}) \\
 &= v \cdot \frac{1 - v^n}{1 - v} = \frac{1}{1+i} \cdot \frac{1 - v^n}{d} = \frac{1}{1+i} \cdot \frac{1 - v^n}{\frac{i}{1+i}}
 \end{aligned}$$

$$a_{\overline{n}|} = \frac{1 - v^n}{i}$$

- the accumulated value (at $t = n$) of an annuity-immediate, where the annual effective rate of interest is i , shall be denoted as $s_{\overline{n}|i}$ and is calculated as follows:

$$\begin{aligned}
 r &= 1+i \\
 s_{\overline{n}|i} &= 1 + (1)(1+i) + \dots + (1)(1+i)^{n-2} + (1)(1+i)^{n-1} \\
 &= \frac{1 - (1+i)^n}{1 - (1+i)} \\
 &= \frac{1 - (1+i)^n}{-i}
 \end{aligned}$$

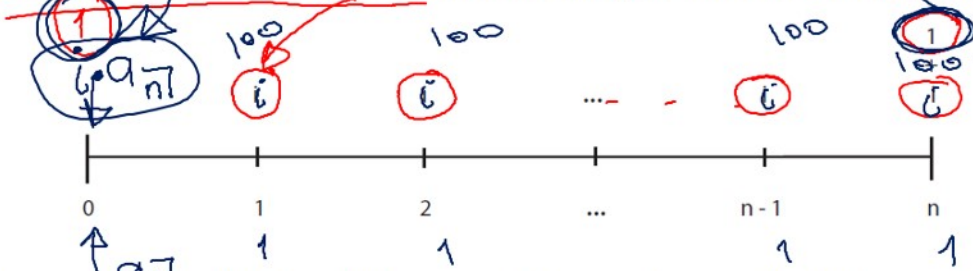
$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}$$

Basic Relationship 1 : $1 = i \cdot a_{\overline{n}|} + v^n$

Consider an n -year investment where 1 is invested at time 0.

The present value of this single payment income stream at $t = 0$ is 1.

Alternatively, consider a n -year investment where 1 is invested at time 0 and produces annual interest payments of $(1) \cdot i$ at the end of each year and then the 1 is refunded at $t = n$.



The present value of this multiple payment income stream at $t = 0$ is $i \cdot a_{\overline{n}|} + (1)v^n$.

Note that $a_{\overline{n}|} \neq \frac{1-v^n}{i} \rightarrow 1 = i \cdot a_{\overline{n}|} + v^n$.

$$i \cdot a_{\overline{n}|} = 1 - v^n$$

$$v^n + i \cdot a_{\overline{n}|} = 1 //$$

Therefore, the present value of both investment opportunities are equal.

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Basic Relationship 2 : $PV(1+i)^n = FV$ and $PV = FV \cdot v^n$

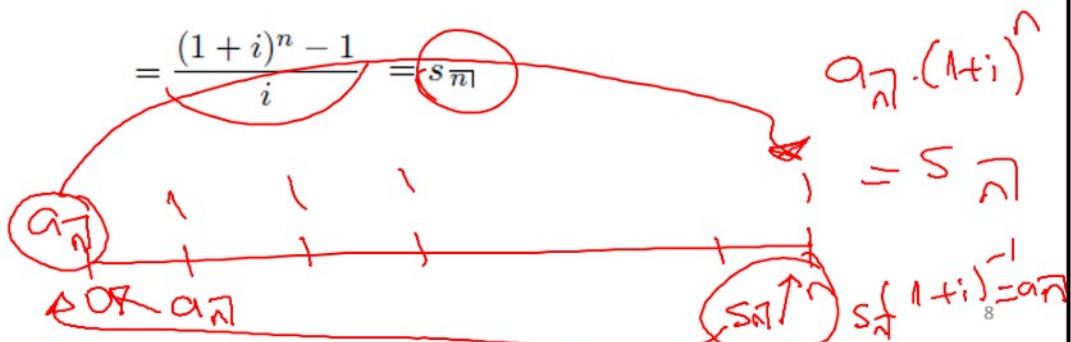
if the future value at time n , $s_{\overline{n}|}$, is discounted back to time 0, then you will have its present value, $a_{\overline{n}|}$

$$s_{\overline{n}|} \cdot v^n = \left[\frac{(1+i)^n - 1}{i} \right] \cdot v^n = \frac{(1+i)^n \cdot v^n - v^n}{i} = \frac{1 - v^n}{i} = a_{\overline{n}|}$$

if the present value at time 0, $a_{\overline{n}|}$, is accumulated forward to time n , then you will have its future value, $s_{\overline{n}|}$

$$a_{\overline{n}|} \cdot (1+i)^n = \left[\frac{1-v^n}{i} \right] (1+i)^n = \frac{(1+i)^n - v^n(1+i)^n}{i}$$

$$= \frac{(1+i)^n - 1}{i} = s_{\overline{n}|}$$



Basic Relationship 3:

$$\frac{1}{a_{\overline{n}|i}} = \frac{1}{s_{\overline{n}|i}} + i$$

Consider a loan of 1, to be paid back over n years with equal annual payments of P made at the end of each year. An annual effective rate of interest, i , is used. The present value of this single payment loan must be equal to the present value of the multiple payment income stream.

$$P \cdot a_{\overline{n}|i} = 1 \Rightarrow P = \frac{1}{a_{\overline{n}|i}}$$

$$P \cdot a_{\overline{n}|i} = 1$$

Alternatively, consider a loan of 1, where the annual interest due on the loan, $(1)i$, is paid at the end of each year for n years and the loan amount is paid back at time n .

In order to produce the loan amount at time n , annual payments at the end of each year, for n years, will be made into an account that credits interest at an annual effective rate of interest i .

The future value of the multiple deposit income stream must equal the future value of the single payment, which is the loan of 1.

$$D \cdot s_{\overline{n}|i} = 1$$

$$D = \frac{1}{s_{\overline{n}|i}}$$

Original

$$D \cdot s_{\overline{n}|i} = 1$$

$$D = \frac{1}{s_{\overline{n}|i}}$$

Interest Repayment Options

Given a loan of 1, there are 3 options in repaying back the loan over the next n years:

Option 1: Pay back the loan and all interest due at time n .

$$\begin{aligned} \text{Total Interest Paid} &= A(n) - A(0) \\ &= \text{Loan} \times (1+i)^n - \text{Loan} \\ &= \text{Loan} \times [(1+i)^n - 1] \end{aligned}$$

$$\text{Loan} \times (1+i)^n$$

Option 2: Pay at the end of each year, the interest that comes due on the loan and then pay back the loan at time n .

$$\begin{aligned} \text{Annual Interest Payment} &= i \cdot \text{Loan} \\ \text{Total Interest Paid} &= i \cdot \text{Loan} \times n \\ &= \text{Loan} \times (i \cdot n) \end{aligned}$$

$$P \cdot a_{\overline{n}|i} = \text{Loan}$$

Option 3: Pay a level annual amount at the end of each year for the next n years.

$$P = \text{Annual Payment} = \frac{\text{Loan}}{a_{\overline{n}|i}}$$

$$\text{Loan} \times (1+i)^n$$

Example 3.1 Find the present value of an annuity which pays \$500 at the end of each half-year for 20 years if the rate of interest is 9% convertible semiannually.

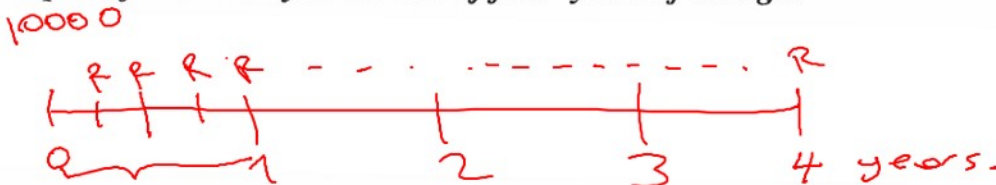
The answer is

$$500 a_{\overline{40}|.045}$$

This can be evaluated using formula (3.2) as

$$500 \frac{1 - (1.045)^{-40}}{.045} = 500(18.40158) = \$9,200.79.$$

Example 3.2 If a college freshman invests a \$10,000 gift at 8% per annum convertible quarterly, how much can be withdrawn at the end of every quarter to use up the fund exactly at the end of four years of college?



Let R be the amount of each withdrawal. The equation of value at the date of investment is

$$10,000 = R a_{\overline{16}|.02} = \frac{1 - (1.02)^{-16}}{0.02} R a_{\overline{16}|.02} = 10,000.$$

Thus, we have

$$R = \frac{10,000}{a_{\overline{16}|.02}} = \frac{10,000}{13.57771} = \$736.50.$$

of payments

$$= 4 \times 4 = 16.$$

$$i^{(4)} = 0.08$$

$$\frac{i^{(4)}}{4} = \frac{0.08}{4} = 0.02 //$$

$$n = 16 \times m$$

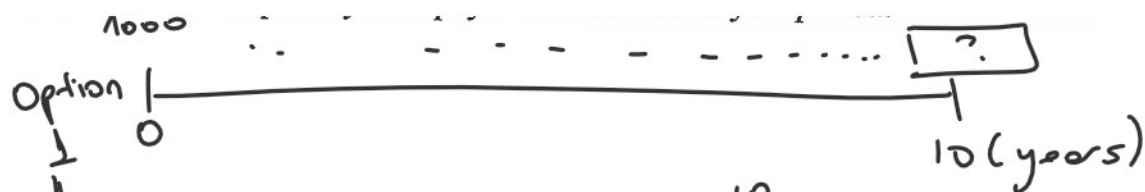
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Example 3.3 Compare the total amount of interest that would be paid on a \$1000 loan over a 10-year period, if the effective rate of interest is 9% per annum, under the following three repayment methods:

- (1) The entire loan plus accumulated interest is paid in one lump-sum at the end of 10 years.
- (2) Interest is paid each year as accrued and the principal is repaid at the end of 10 years.
- (3) The loan is repaid by level payments over the 10-year period.

$$i = 0.09$$

Option | ————— | ?



(1) $A.V = 1000 \cdot (1 + 0.09)^{10}$

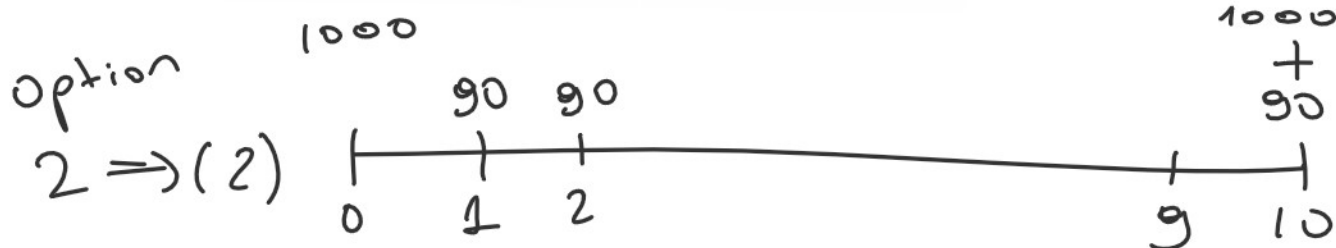
Total interest Amount = $A.V - 1000$

1. The accumulated value of the loan at the end of 10 years is

$$1000(1.09)^{10} = 2367.36$$

Thus, the total amount of interest paid is equal to

$$2367.36 - 1000.00 = 1367.36.$$



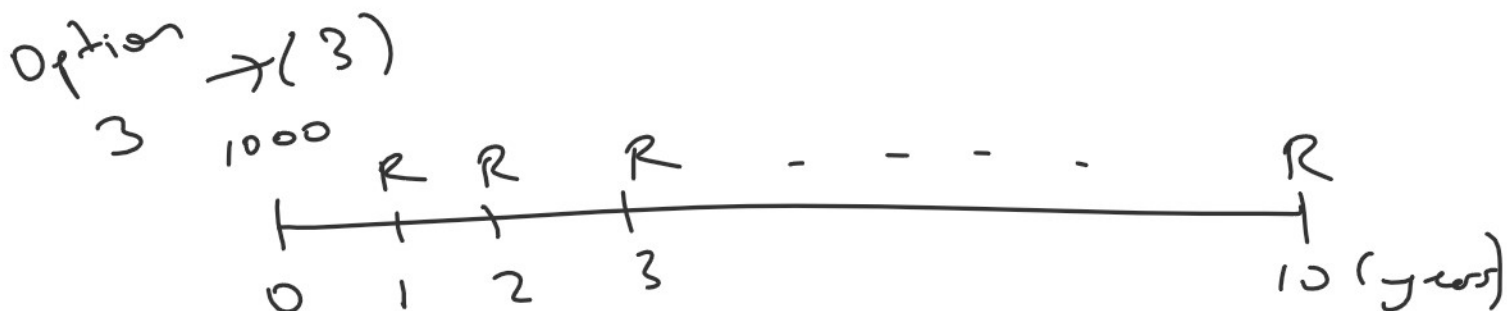
$$i = 0.09$$

$$1000 \cdot (0.09) = 90$$

$$90 \times 10 = 900 // = T.I.$$

2. Each year the loan accrues interest of $1000(0.09) = \$90$, so that the total amount of interest paid is equal to

$$10 \cdot 90 = \$900.00.$$



$$R \cdot a_{\overline{10}|0.09} = 1000$$

$$R \cdot a_{\overline{10}|0.09} = 1000$$

$$\frac{1-v^n}{i} = a_{\overline{n}|i}$$

$$a_{\overline{10}|0.09} = \frac{1-(1.09)^{-10}}{0.09}$$

3. Let the level payment be R . An equation of value for R at the inception of the loan is

$$R a_{\overline{10}|} = 1000$$

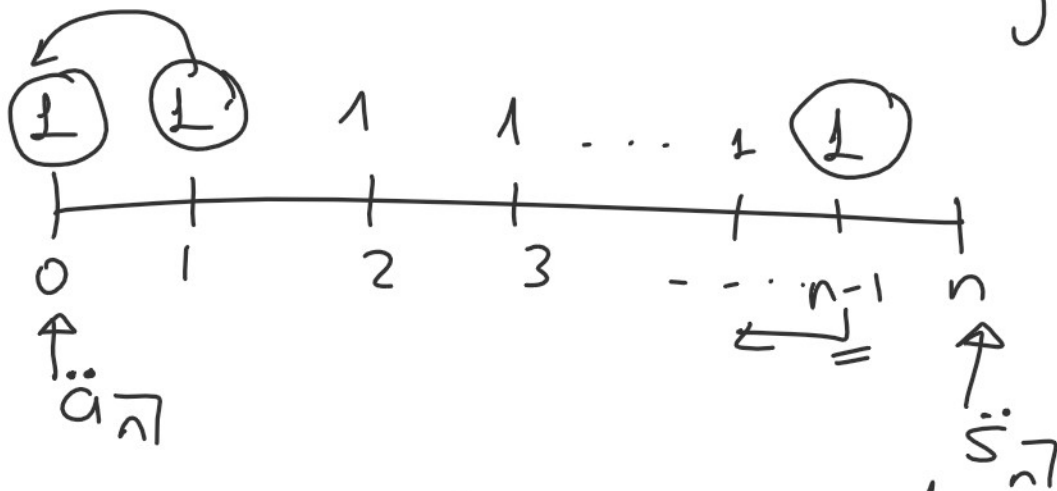
which gives

$$R = \frac{1000}{a_{\overline{10}|}} = \frac{1000}{6.417658} = 155.82 //$$

Thus, the total amount of interest paid is equal to

$$10(155.82) - 1000 = \$558.20 \approx \text{T. 1. around.}$$

Annuity - due $\Rightarrow$ at the beginning.



$$\ddot{a}_{\overline{n}|} = 1 + 1.v + 1.v^2 + \dots + 1.v^{n-1}$$

$$= \frac{1-v^n}{1-v} = \boxed{\frac{1-v^n}{d}} = \ddot{a}_{\overline{n}|}$$

PV
annuity
due

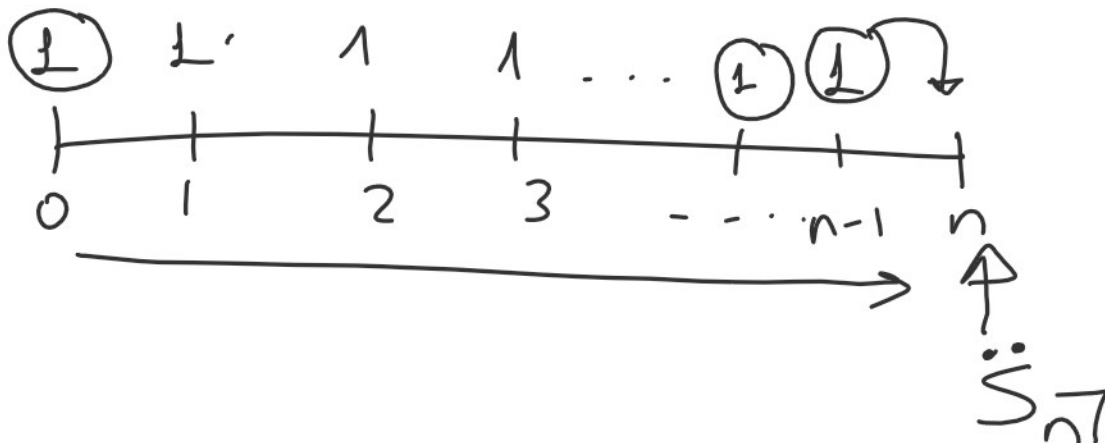
$$\boxed{a_{\overline{n}|} = \frac{1-v^n}{i}}$$

annuity
immediate

$$|u_n| = \boxed{i}$$

1 more tick

$$\begin{aligned}\ddot{a}_{\overline{n}|i} &= 1 + (1)v + (1)v^2 + \dots + (1)v^{n-2} + (1)v^{n-1} \\ &= \frac{1 - v^n}{1 - v} \\ &= \frac{1 - v^n}{d}\end{aligned}$$



$$\begin{aligned}\ddot{S}_{\overline{n}|i} &= 1 \cdot (1+i) + 1 \cdot (1+i)^2 + \dots + 1 \cdot (1+i)^n \\ &= (1+i) \left[1 + (1+i) + \dots + (1+i)^{n-1} \right] \\ &= (1+i) \frac{1 - (1+i)^n}{1 - (1+i)}\end{aligned}$$

$$\begin{aligned}\text{d: } \frac{i}{1+i} &= \frac{1 - (1+i)^n}{\frac{-i}{1+i}} = \boxed{\frac{(1+i)^n - 1}{d}} = \ddot{S}_{\overline{n}|i}\end{aligned}$$

F.V. of ann-due.

$$\boxed{S_{\overline{n}|i} = \frac{(1+i)^n - 1}{i}}$$

F.V. of ann-imn

the accumulated value (at $t = n$) of an annuity-due, where the annual effective rate of interest is i , shall be denoted as $\ddot{s}_{\overline{n}|i}$ and is calculated as follows:

$$\begin{aligned}
 \ddot{s}_{\overline{n}|i} &= (1)(1+i) + (1)(1+i)^2 + \cdots + (1)(1+i)^{n-1} + (1)(1+i)^n \\
 &= (1+i)[1 + (1+i) + \cdots + (1+i)^{n-2} + (1+i)^{n-1}] \\
 &= (1+i) \left[\frac{1 - (1+i)^n}{1 - (1+i)} \right] \\
 &= (1+i) \left[\frac{1 - (1+i)^n}{-i} \right] \\
 &= (1+i) \left[\frac{(1+i)^n - 1}{i} \right] \\
 &= \frac{(1+i)^n - 1}{d}
 \end{aligned}$$

4

Basic Relationship 3: $\frac{1}{\ddot{a}_{\overline{n}|i}} = \frac{1}{\ddot{s}_{\overline{n}|i}} + d$

Consider a loan of 1, to be paid back over n years with equal annual payments of P made at the beginning of each year. An annual effective rate of interest, i , is used. The present value of the single payment loan must be equal to the present value of the multiple payment stream.

$$P \cdot \ddot{a}_{\overline{n}|i} = 1 \Rightarrow P = \frac{1}{\ddot{a}_{\overline{n}|i}} \quad P \cdot \ddot{a}_{\overline{n}|i} = 1$$

Alternatively, consider a loan of 1, where the annual interest due on the loan, $(1) \cdot d$, is paid at the beginning of each year for n years and the loan amount is paid back at time n .

In order to produce the loan amount at time n , annual payments at the beginning of each year, for n years, will be made into an account that credits interest at an annual effective rate of interest i .

The future value of the multiple deposit income stream must equal the future value of the single payment, which is the loan of 1.

$$D \cdot \ddot{s}_{\overline{n}|i} = 1 \Rightarrow D = \frac{1}{\ddot{s}_{\overline{n}|i}} \quad D \cdot \ddot{s}_{\overline{n}|i} = 1$$

The total annual payment will be the interest payment and account payment: $d + \frac{1}{\ddot{s}_{\overline{n}|i}}$

- Considerable confusion has often been created by treating the annuity-immediate and the annuity-due as if they were drastically different. In reality, they refer to exactly the same series of payments evaluated at different points in time.

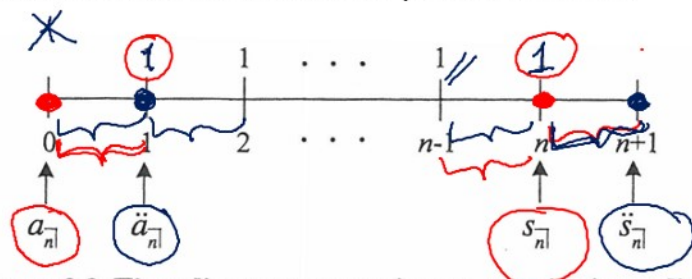


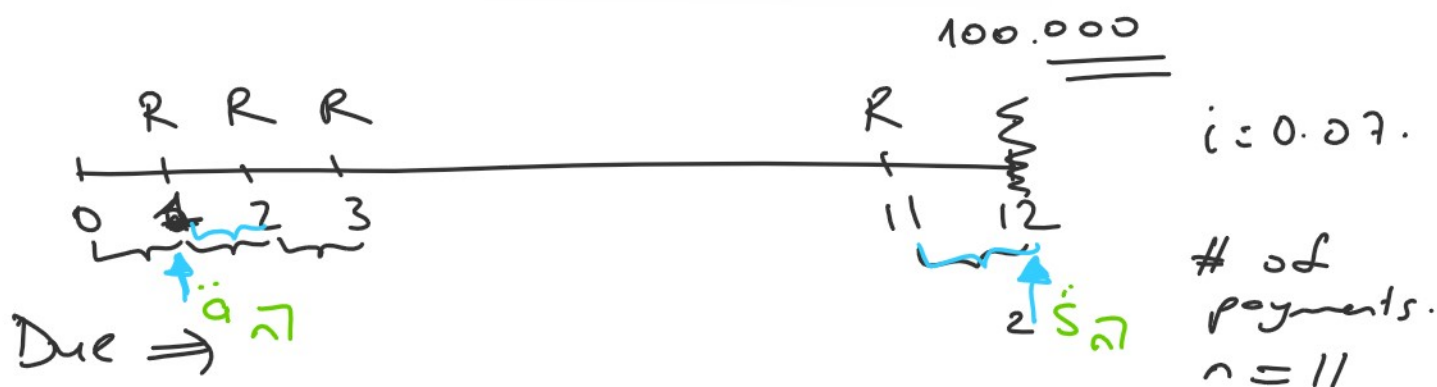
Figure 3.3 Time diagram comparing an annuity-immediate

with an annuity-due

$$a_n \cdot (1+i) = \ddot{a}_n \quad s_n \cdot (1+i) = \ddot{s}_n$$

of payments = n.

Example 3.4 An investor wishes to accumulate \$100,000 in a fund for retirement at the end of 12 years. To accomplish this the investor plans to make deposits at the end of each year, the final payment to be made one year prior to the end of the investment period. How large should each deposit be if the fund earns 7% effective?



comparison date => $t = 12$.

$$R \cdot \ddot{s}_{\overline{11}|i} = 100000$$

$$\ddot{s}_n = \frac{(1+i)^n - 1}{i}$$

This problem should be read carefully. There will be only 11 payments, not 12. Since we are interested in the accumulated value one year after the last payment, the equation of value is

This problem should be read carefully. There will be only 11 payments, not 12. Since we are interested in the accumulated value one year after the last payment, the equation of value is

d

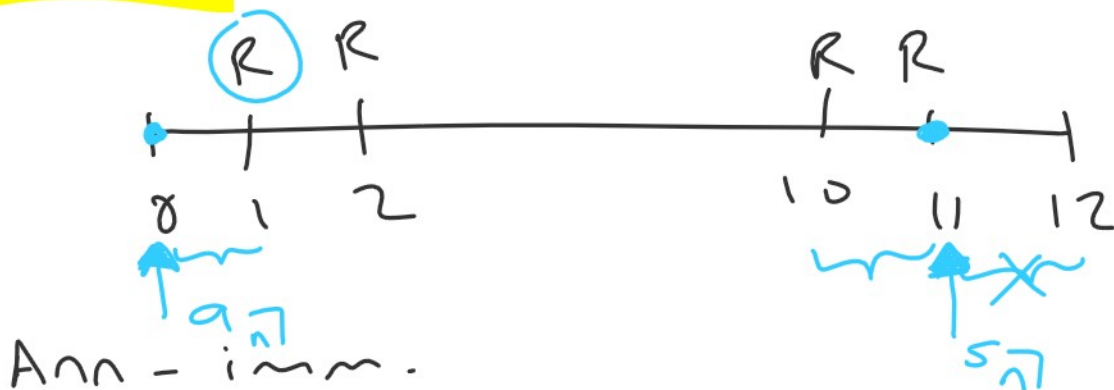
$$R \ddot{s}_{\overline{11}|} = 100,000$$

where R is the annual deposit. Solving for R we have

$$R = \frac{100,000}{\ddot{s}_{\overline{11}|}} = \frac{100,000}{16.88845} = \$5921.21.$$

100.0000

n = 11



comp. date $t = 11$

$$R \cdot S_{\overline{11}|} = 100000 \cdot (1+i)^{-1} \text{ where } i = 0.07$$

con date $t = 12$

$$R \cdot S_{\overline{11}|} \cdot (1+i) = 100000.$$